

Population Shifts in Helsinki 2003–2023

Prediction, Explanation & Visualization

Machine Learning Analysis of Helsinki District-Level Demographic Data

Dataset: Population2003-2023.xlsx

Source: Helsinki Statistics • Avoindata.fi

Scope: Helsinki areas, districts, and major districts

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Abstract

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Title of the thesis Population Shifts in Helsinki – 2003-2023		
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Abstract In this project present a cloud-based platform that allows users to write and execute code directly from a web browser. The system supports multiple programming languages and focuses on secure execution using Docker containers. Since running user-submitted code can be risky, the platform ensures isolation and protection using containerization and security techniques. The solution also follows modern cloud service models such as SaaS and PaaS. Overall, the project demonstrates how different technologies can be combined to build a practical and secure real-world application.		
Keywords Cloud, Docker, Security, Code Execution, HTML, CSS, Java, Localhost, Python.		

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1 Summary

This project analyses district-level population change in Helsinki from 2003 to 2023 using official demographic statistics from Helsinki Statistics (Avoindata.fi). The original assignment focused on Finnish national population trends; however, the available dataset is specific to Helsinki, so the project scope was adjusted accordingly.

Helsinki grew by 104,312 people over the study period. Natural increase (births minus deaths) contributed 28,225 people, while total net migration contributed 75,155 people, confirming that migration is the dominant driver of population growth in Helsinki.

a - Key Headline Figures

Total growth: +104,312 people • Natural increase: +28,225 • Net migration: +75,155
• Study period: 2003–2023 (19 years, 2017 excluded) • Areas analysed: 204 unique area names • Dataset rows (after cleaning): ~3,400

Four machine learning techniques were applied:

- Linear Regression — stable long-term population forecasting
- Polynomial Regression — model comparison and fit evaluation
- K-Means Clustering — grouping areas by demographic growth pattern
- Isolation Forest — detecting years with unusual population shifts

I – Dataset Overview

The dataset (Population2003-2023.xlsx) is an Excel workbook with one sheet per year covering Helsinki area and district-level population change statistics from 2003 to 2023. Each sheet originally contains Finnish-language column headers with formatting and metadata rows that must be removed before analysis.

I.I – Scope

Attribute	Value
File	Population2003-2023.xlsx
Geographic scope	Helsinki areas, districts, and major districts
Time period	2003–2023 (year 2017 missing — excluded)
Years available	19 years
Unique area names (after cleaning)	204
Total rows after cleaning	~3400
Source	Helsinki Statistics • Avoindata.fi

I.II – Variables

Name	Finnish original	Description
Area	Alue	Helsinki area, district or major district name
Population_Change	Väestönmuutos	Net change in population during the year
Births	Syntyneet	Number of live births
Deaths	Kuolleet	Number of deaths
Natural_Increase	Luonnollinen väestönlisäys	Births minus deaths
Total_Migration_In	Muuttaneet yhteensä	All arrivals (domestic + international)
Total_Migration_Out	Muuttaneet yhteensä	All departures (domestic + international)
Total_Net_Migration	Nettomuutto	Migration In minus Migration Out
External_Migration_In/Out/Net	Ulkomainen muutto	Cross-border (international) migration
Internal_Migration_In/Out/Net	Kuntien välinen muutto	Domestic (within-Finland) migration
Year	(added)	Calendar year — appended during processing

2 - Limitations

The project focuses on Helsinki rather than the whole of Finland because the available dataset is Helsinki-specific. Results should not be interpreted as representing all Finnish municipalities.

Limitation	Impact
Year 2017 missing from source data	Excluded entirely; analyses cover 19 instead of 21 years
No age-group or sex/gender breakdown	Direct aging-rate analysis is not possible
Data starts at 2003	Pre-2003 baseline trends are unavailable
Helsinki only — not national	Results cannot be generalised to all of Finland
Areas labelled 'Muut' are catch-all admin rows	Excluded from area-level analyses to avoid distortion

3 - Data Preparation

Raw data required significant preparation before analysis could begin. Each yearly sheet had multi-level Finnish-language headers, formatting rows, and metadata rows (source references, blank spacer rows) mixed with the actual data rows.

I – Data Loading

The workbook is opened with pandas ExcelFile. Only sheets with purely numeric names (i.e. year sheets) are retained. This filters out any summary or metadata sheets.

Code:

```
load_data = 'Population2003-2023.xlsx'

excel_file = pd.ExcelFile(load_data)

year_sheets = [s for s in excel_file.sheet_names if s.isdigit()]

# Result: 19 year sheets (2003-2023, excluding 2017)
```

II - Combining Yearly Sheets

The actual data row anchor is located dynamically by searching for the string 'Koko Helsinki' (Whole Helsinki) within each sheet. Data is read from that row onwards, trimmed to 14 columns, and renamed to English. A Year column is appended before all yearly frames are concatenated into a single dataframe.

Code:

```
column_names = [
    'Area', 'Population_Change', 'Births', 'Deaths',
    'Natural_Increase', 'Total_Migration_In', 'Total_Migration_Out',
    'Total_Net_Migration', 'External_Migration_In', 'External_Mi-
    gration_Out',
    'External_Net_Migration', 'Internal_Migration_In',
    'Internal_Migration_Out', 'Internal_Net_Migration'
]

# Each sheet: locate 'Koko Helsinki' row, slice from there, rename, add Year
data_raw = pd.concat(all_years_data, ignore_index=True)
```

III - Cleaning Steps

The following cleaning steps were applied in sequence. Each step was necessary to produce a dataset suitable for visualisation and machine learning.

Step	Action	Reason
1	Drop rows where Area is null or blank	Remove metadata and spacer rows
2	Strip whitespace from Area strings	Prevent duplicate-looking area names

3	Cast all numeric columns with <code>pd.to_numeric(errors='coerce')</code>	Non-numeric values become NaN instead of raising an error
4	Exclude rows/areas containing 'Muut'	Remove administrative catch-all categories
5	Exclude areas with special characters (* or))	Remove footnote and aggregate rows
6	Require ≥ 5 valid years per area for modelling	Ensure enough data points for regression

After cleaning: ~3,400 rows, 204 unique area names, 19 years.

4 - Exploratory Analysis & Visualisation

I - Helsinki-Level Population Change Trend

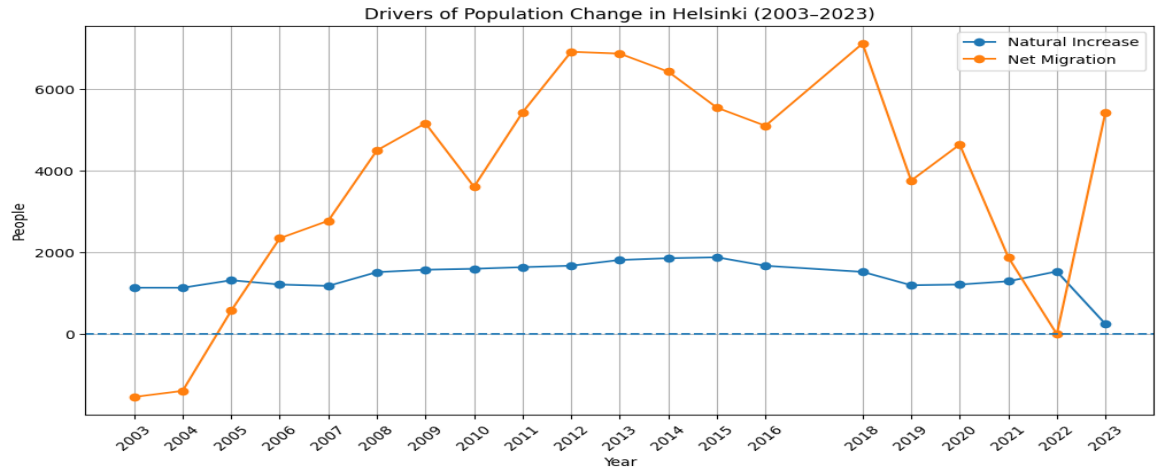
The first visualisation plots total Helsinki (Koko Helsinki) annual population change from 2003 to 2023. The early years 2003–2004 show negative change. Growth became consistently positive after 2005, with notable fluctuation in the 2019–2022 period.

II - Drivers of Population Change

A dual-line chart compares Natural Increase (births minus deaths) with Total Net Migration over time.

Driver	Trend	Key Observation
Natural Increase	Declining, recently negative	Deaths now outnumber births in Helsinki. Natural increase has contributed less each year.
Total Net Migration	Volatile — peaked 2012–2018, dropped 2019–2021, surged 2023	Migration is the dominant and most variable driver. Even short spikes offset years of natural decline.

Key Insight - 'Migration is unstable but powerful.' Net migration alone accounts for 72% of total Helsinki population growth over the study period (75,155 of 104,312 people).

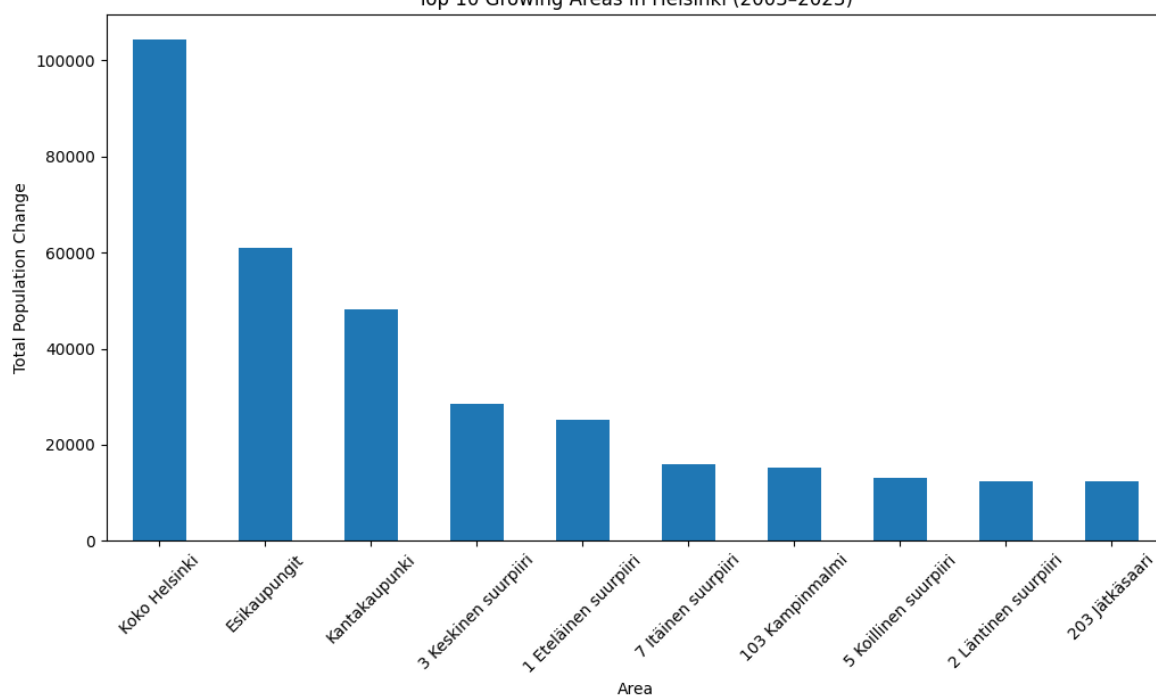


III - Top Growing & Declining Areas

Total population change is summed across all years per area. 'Muut' categories are excluded. The top 10 growing and top 10 declining areas are identified and visualised as bar charts.

- Growth is concentrated in newer development zones: Jätkäsaari, Kalasatama, Kampinmalmi, Kallio, and central redevelopment areas.
- Some older or peripheral residential areas show flat or negative cumulative change, indicating internal migration from suburbs toward the city centre.

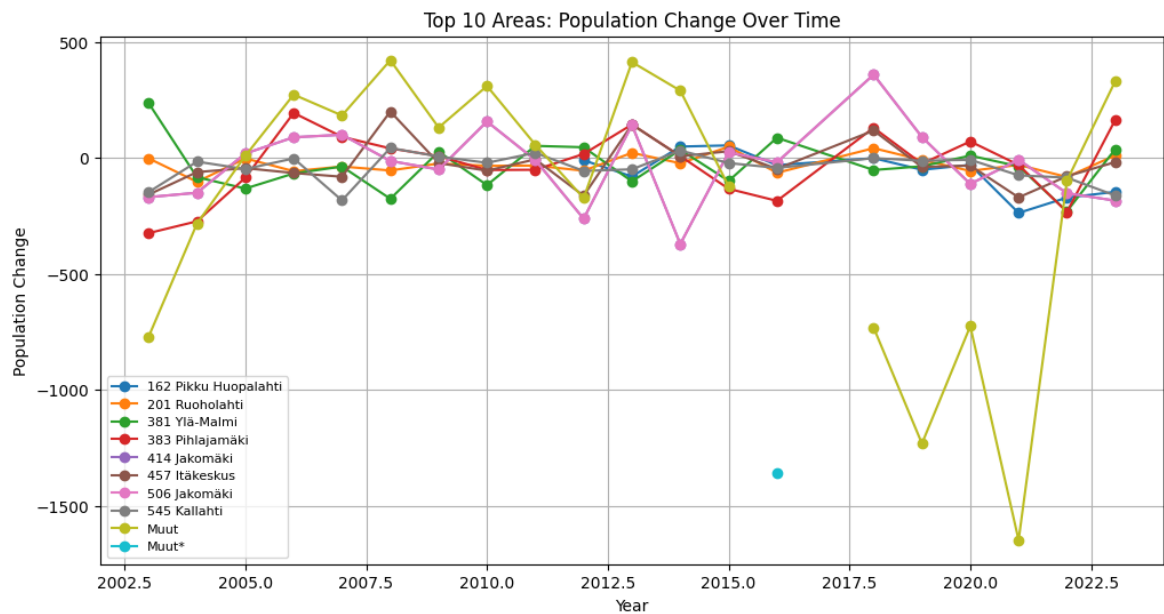
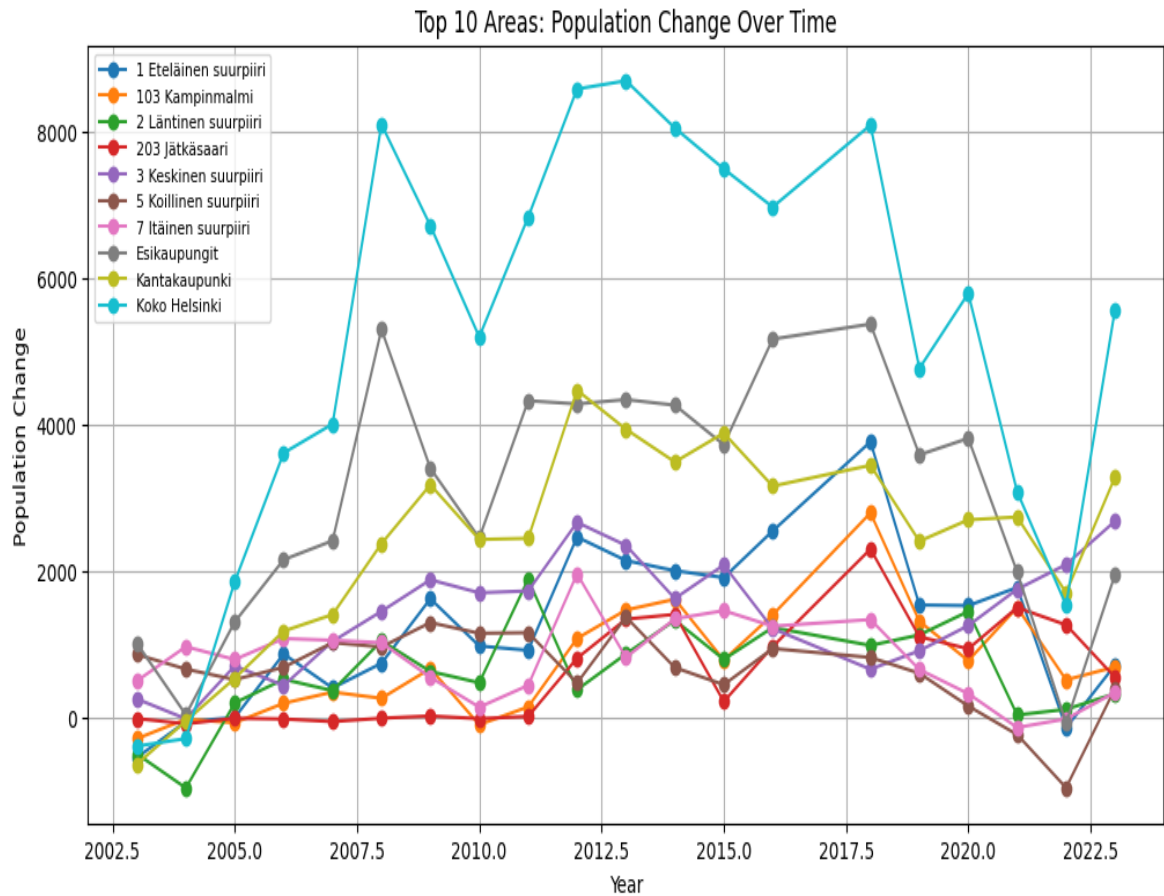
Top 10 Growing Areas in Helsinki (2003–2023)



IV - Growth Type Classification

Each area is assigned a Growth Type label based on which driver was larger over the full period.

Growth Type	Condition	Interpretation
Migration-driven	Total_Net_Migration > Natural_Increase	Growth is primarily from people moving in
Natural-growth	Natural_Increase > Total_Net_Migration	More births than deaths drives the increase
Balanced	Equal contributions	Neither driver dominates



5. Forecasting Models (2024–2040)

Two regression models were trained on the 'Koko Helsinki' series to forecast annual population change from 2024 to 2040. Year is used as the sole predictor feature.

I - Linear Regression

A standard Ordinary Least Squares linear regression is fitted on all available historical years. It models population change as a straight-line function of time and extrapolates this trend into the future.

Code -

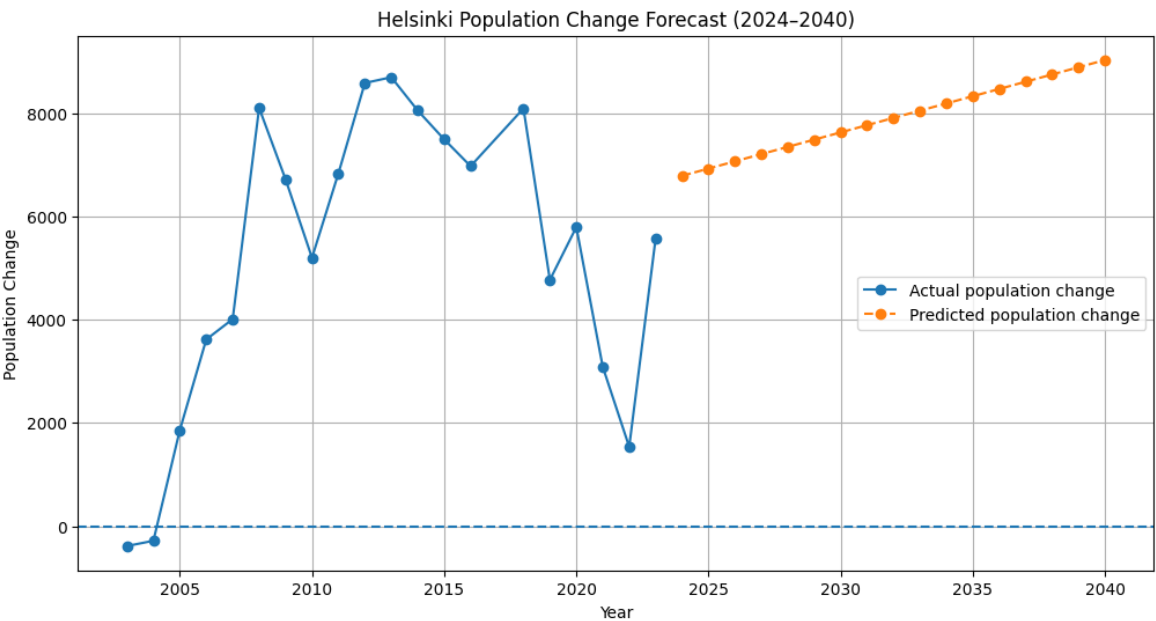
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(helsinki[['Year']], helsinki['Population_Change'])

future_years = pd.DataFrame({'Year': range(2024, 2041)})

predictions = model.predict(future_years)
```



II - Polynomial Regression (Degree 2)

A degree-2 polynomial adds a squared Year term to capture non-linear patterns in the historical data. This reduces training error but introduces curvature that can become unrealistic when extended into the future.

Code -

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error

poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(helsinki[['Year']])
poly_model = LinearRegression()
poly_model.fit(X_poly, y)
```

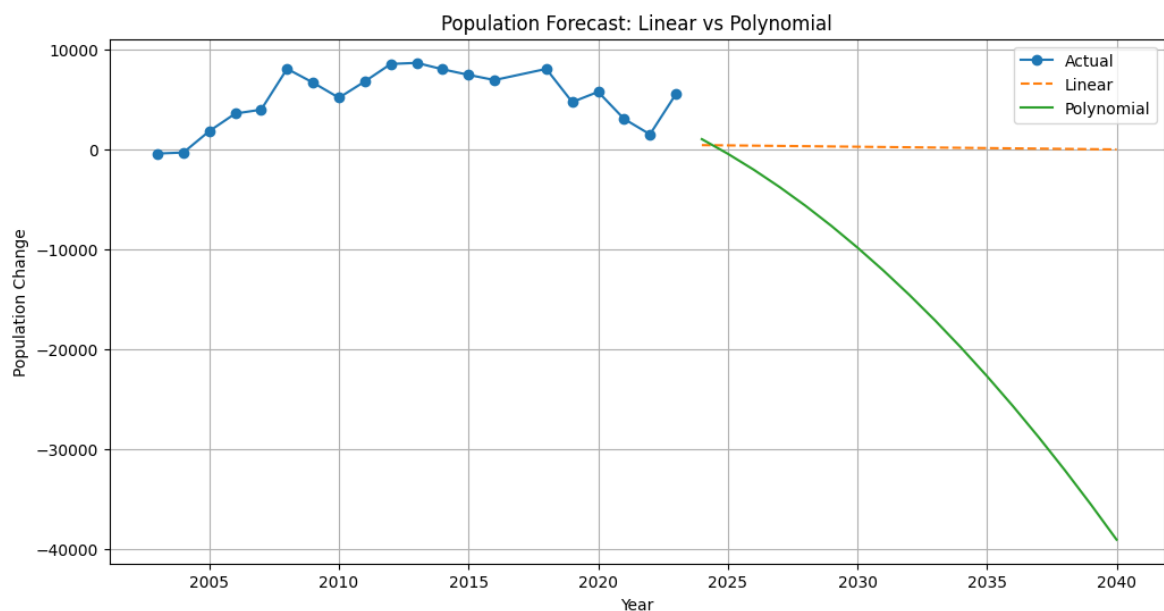
Model	Training MSE	Future Trend	Chosen for Fore- cast?
Linear Regression	7,103,549	Stable, gradual up- ward slope	Yes

Polynomial Regression (deg. 2)	1,739,268	Drops sharply after ~2030 — unrealistic	No
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Model Decision - Polynomial regression fits historical data better (lower MSE) because Helsinki population change is not a straight line. However, the curvature extrapolates into an implausible sharp drop. Linear regression is selected for forecasting because it produces stable, interpretable, and policy-relevant long-term estimates. Model accuracy on past data does not equal reliability for the future.

III - Area-Level Forecasts

The same linear model is applied independently to each area with at least five valid annual observations. This generates a per-area forecast table for 2024–2040. Areas with insufficient data are skipped. Results are summarised by sorting total predicted 2024–2040 change per area.



6. Clustering Analysis

K-Means clustering groups Helsinki areas by demographic similarity, revealing structural patterns that are invisible when looking at individual areas in isolation.

I - Feature Set & Preprocessing

Three aggregated features per area are used: Population Change, Natural_Increase, and Total_Net_Migration (summed over all years). Areas with 'Muut', asterisks, or parentheses in their name are excluded. All features are standardised with StandardScaler before clustering so that no single variable dominates due to scale differences.

```
Code - from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaled_data = scaler.fit_transform(area_analysis[features])
```

II - Cluster Model

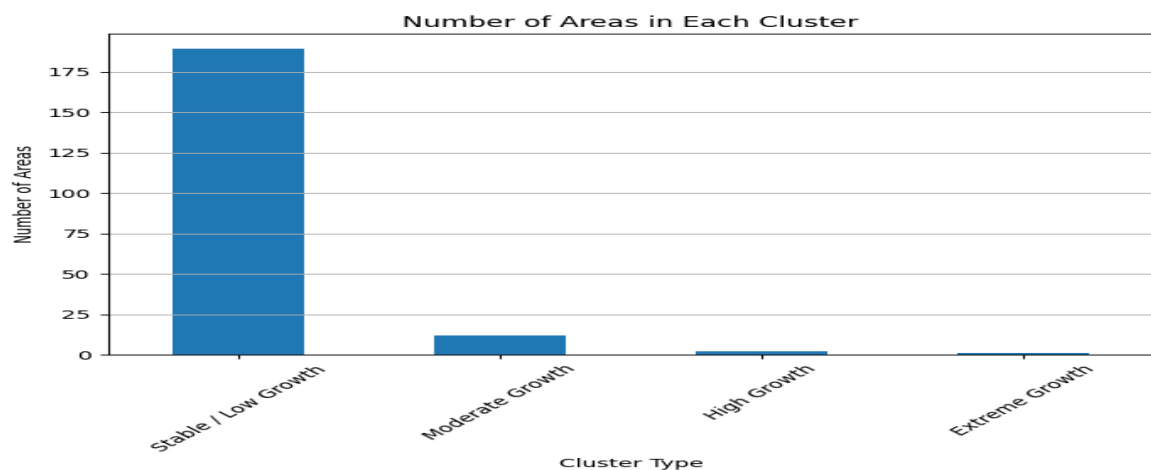
An initial K-Means model with k=3 is fitted to explore broad area groupings. The three clusters loosely map to high-growth, stable, and declining areas. This provides a useful overview but lacks granularity.

III - Cluster Model (Final)

A four-cluster model provides finer resolution and clearer interpretation. Cluster labels are assigned manually based on cluster mean values for the three features.

Cluster ID	Label	Avg. Pop. Change	Dominant Driver	Example Areas
1	Extreme Growth	~104,312	Very strong net migration	Koko Helsinki (total), major aggregates
3	High Growth	~54,517	Strong net migration	Kantakaupunki, Esikaupungit
0	Moderate Growth	~14,101	Mix of migration & natural	Jätkäsaari, Kalasatama, central districts
2	Stable / Low Growth	~838	Weak or negligible	Most individual sub-districts

Clustering Insight - Only a small number of areas have very strong growth. The majority of Helsinki sub-districts are Stable or Low Growth. In all high-growth clusters, net migration is larger than natural increase again confirming migration as the primary population driver.



7. Anomaly Detection

I - Model: Isolation Forest

An Isolation Forest model is applied to Helsinki's annual Population Change series to detect years where the value deviates significantly from the overall trend. Isolation Forest is appropriate here because population change is not normally distributed and does not follow a simple parametric shape.

Two contamination parameters are tested: 0.15 (broader detection, ~3 anomalies) and 0.05 (stricter detection). Anomaly == -1 in the model output indicates an outlier year.

```
Code - from sklearn.ensemble import IsolationForest

model = IsolationForest(contamination=0.15, random_state=42)

helsinki['Anomaly'] = model.fit_predict(helsinki[['Population_Change']])

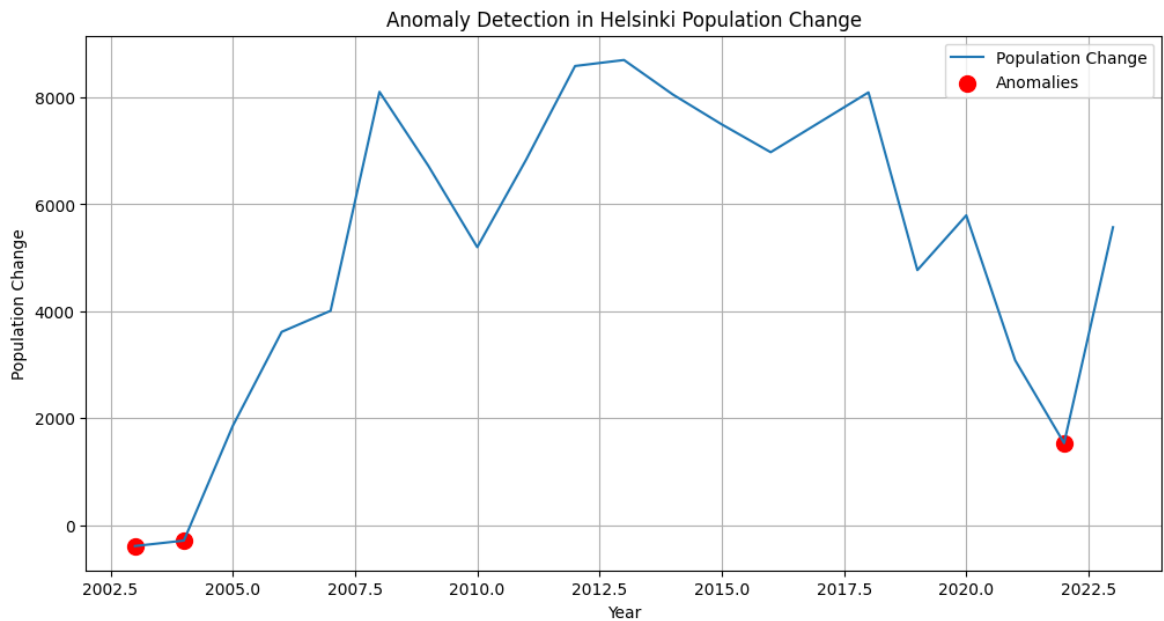
# Anomaly == -1 → outlier year

anomalies = helsinki[helsinki['Anomaly'] == -1]
```

II - Anomalous Years Detected

Year	Population Change	Why Anomalous
2003	Negative	Early-period negative growth; very different from later trend
2004	Negative / very low	Continued negative growth before the sustained positive trend began
2022	Much lower than surrounding years	Likely influenced by post-pandemic disruption to migration patterns

The anomaly chart overlays red scatter markers on the time-series line for flagged years. This is a useful diagnostic tool for identifying years that warrant deeper investigation for example, checking whether economic conditions, national policy changes, or global events (such as the COVID-19 pandemic) explain the deviation.



8. Libraries & Dependencies

Library	Version (typical)	Purpose
Pandas	2.x	Data loading, cleaning, reshaping, groupby aggregation
Numpy	1.x / 2.x	Numerical operations, NaN handling
Matplotlib	3.x	Line charts, bar charts, scatter plots
Seaborn	0.x	Enhanced statistical visualisation styling
scikit-learn	1.x	LinearRegression, PolynomialFeatures, KMeans, IsolationForest, StandardScaler, MSE
openpyxl	3.x	Reading .xlsx files (pandas backend)

9. Key Findings Summary

Topic	Finding
Total growth (2003–2023)	Helsinki grew by +104,312 people over the study period.
Primary driver	Net migration (+75,155) accounts for 72% of total growth; natural increase contributes only 28%.
Natural growth trend	Natural increase is declining and has turned negative in recent years, deaths now exceed births.
Migration pattern	Net migration is volatile: peaked 2012–2018, dropped sharply 2019–2021, surged again in 2023.
Area disparities	New development zones (Jätkäsaari, Kalasatama) grow rapidly; older suburbs show flat or negative change.

Forecasting model	Linear regression chosen over polynomial for forecasting more stable and reliable long-term despite higher MSE.
Clustering	Four area types identified: Extreme Growth, High Growth, Moderate Growth, Stable/Low Growth. Migration dominates in top clusters.
Anomaly years	2003, 2004, and 2022 flagged as statistically unusual by Isolation Forest.

10. Conclusion

Helsinki's population has grown strongly since 2003, but growth is uneven across areas and almost entirely dependent on migration. As natural increase turns negative, migration becomes the sole engine of population growth making the city's future demographic trajectory highly sensitive to immigration policy, housing availability, and global conditions.

The results show that new and central development areas (Jätkäsaari, Kalasatama, Kampinmalmi) will likely continue to absorb the largest population increases, creating pressure on housing, transport, schools, and public services in those zones. Policymakers should account for these spatial imbalances when planning urban infrastructure.

From a modelling perspective, this project demonstrates that machine learning can meaningfully explain and forecast demographic change at the sub-city level. However, model results must be interpreted carefully: lower training error does not guarantee better future predictions (as seen in the polynomial vs. linear comparison), and all forecasts assume past trends continue.

Presentation Summary

I analysed Helsinki population-change data from 2003 to 2023. I cleaned and combined 19 yearly Excel sheets, visualised population trends, and compared natural increase with migration. Results show Helsinki's growth is almost entirely migration driven. I then tested and compared forecasting models, clustered areas by growth pattern using K-Means, and used Isolation Forest to detect unusual years. The key finding: migration is unstable but powerful and it is what keeps Helsinki growing.